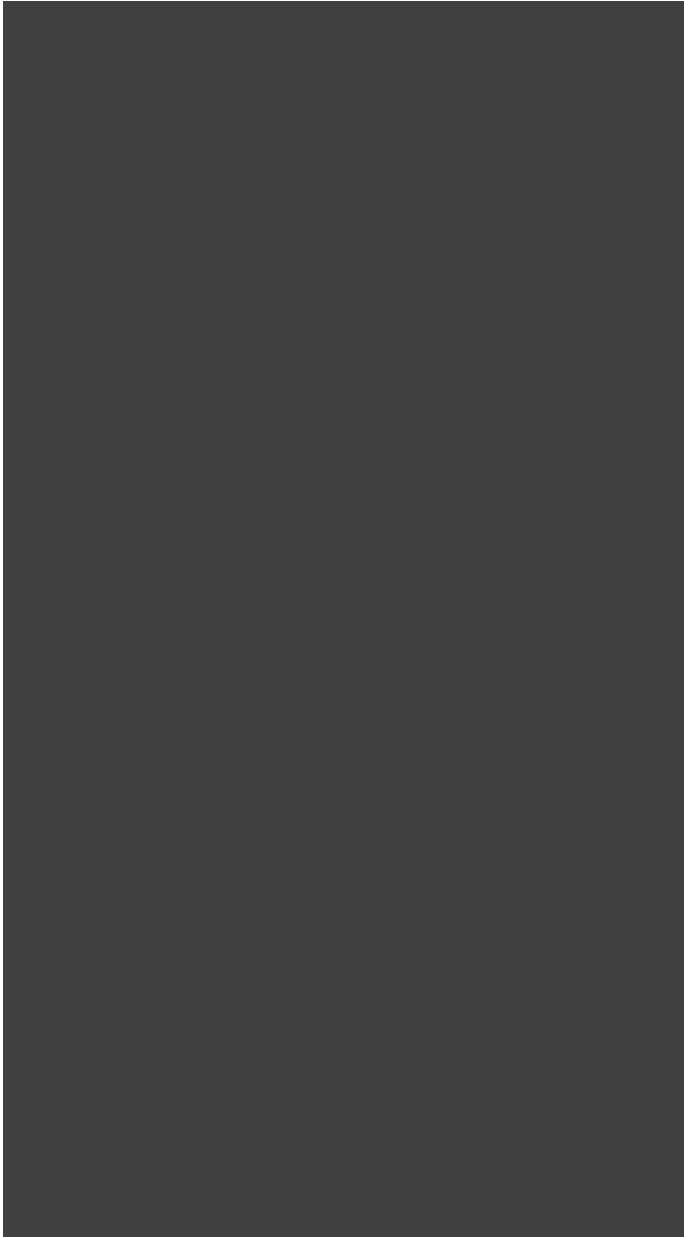




COMPUTING OVER ENCRYPTED DATA: HOMOMORPHIC ENCRYPTION AND CRYPTDB

HOMOMORPHIC ENCRYPTION



Encryption Enc is called homomorphic with respect to an operation $*$ if $Enc(x * y) = Enc(x) * Enc(y)$



That is given encrypted forms of x and y , in order to compute encrypted form of $x * y$ one does not need to decrypt $Enc(x)$ and $Enc(y)$

Computations happen over encrypted values!

PARTIAL VS FULLY HOMOMORPHIC SCHEMES

Partially homomorphic encryption:

with respect just to one operation (+ or $\times$)

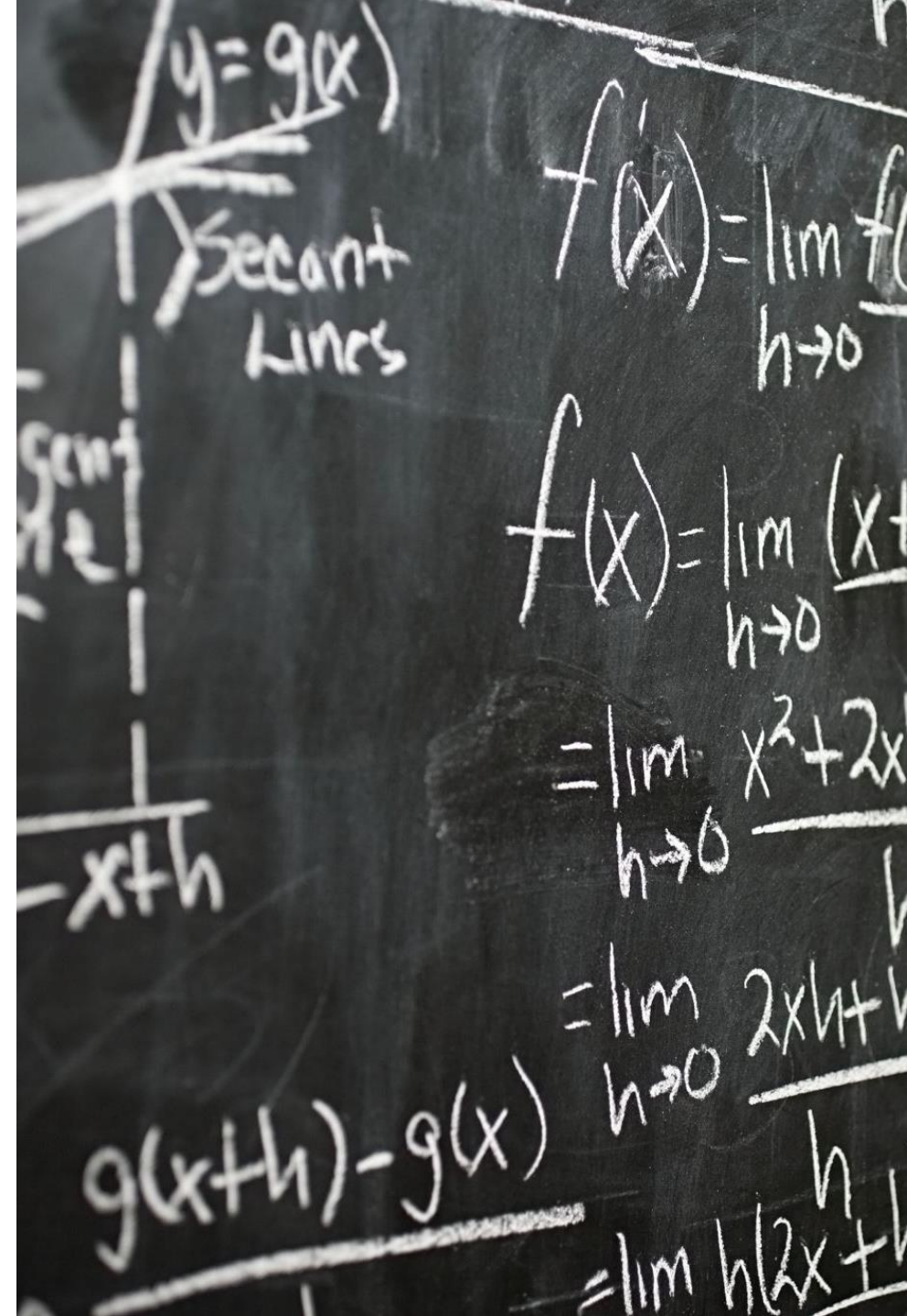
RSA (unpadded) is homomorphic with respect to multiplication

- $Enc(x) = x^e \bmod n$
- $Enc(x) \times Enc(y) = x^e \times y^e = (xy)^e = Enc(xy)$

Other schemes: Paillier (can be used in voting), ...

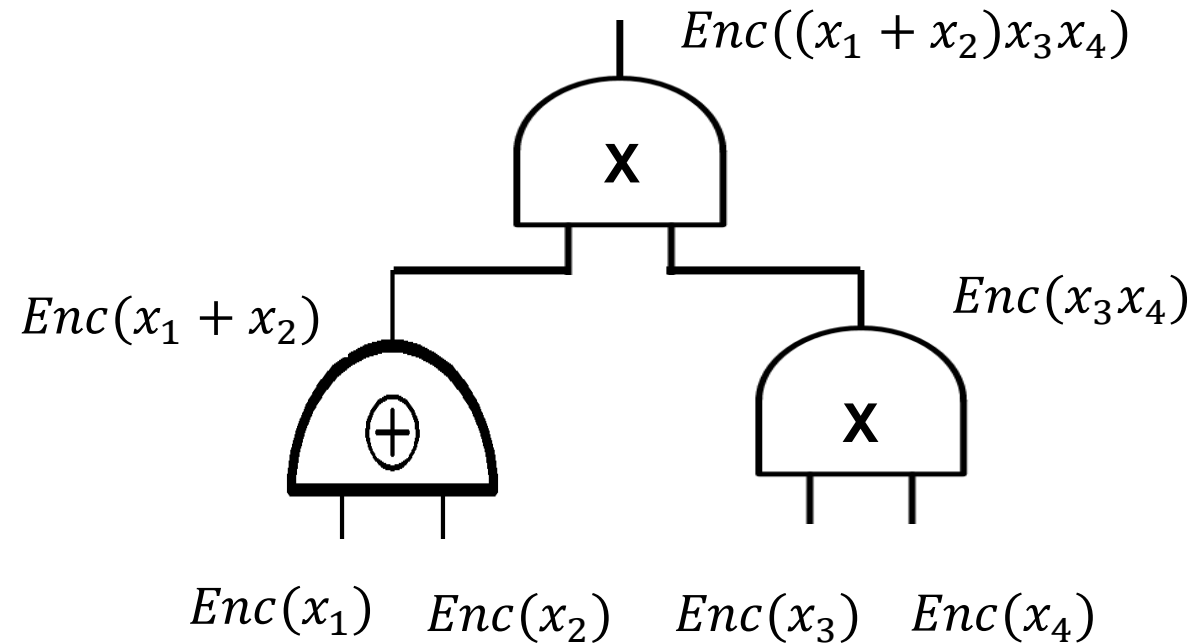
Fully homomorphic schemes [RAD'78]:

- With respect to multiplication and addition
- Allow to perform arbitrary computations
- Existence is by no means obvious



HOW TO COMPUTE ARBITRARY FUNCTIONS

For us, programs = functions = Boolean circuits with XOR ($+ \bmod 2$) and AND ($\times \bmod 2$) gates



Takeaway: If you can compute XOR and AND on encrypted bits, you can compute any program

BREAKTHROUGH: FHE IS POSSIBLE!

Craig Gentry: first fully homomorphic encryption scheme is announced by IBM in 2009
(based on [lattice cryptography](#) and introduced [bootstrapping](#) to control noise growth)

The scheme is impractical for many applications:

- Ciphertext size and computation time increase sharply as one increases the security level
- Operations are **1000x to 1,000,000x** slower than on plaintext
 - Ciphertexts are huge polynomials (message + noise)
 - Noise grows after every operation
 - Bootstrapping resets noise (very expensive)
- Key's size is also an issue

FHE security is based on [Learning With Errors \(LWE\)](#) and this is believed to be **quantum-resistant**

RECENT DEVELOPMENTS

New more efficient schemes and implementations since 2010, key size is reduced at least to 600Kb (~2016)

HELib is an open-source implementation (2013, new version 2018)) (C++)

More implementations available, including in R and Python

New library SEAL made available by Microsoft in 2018

Others: PALISADE, OpenFHE, HEAAN, FHEW, TFHE, LattiGo ...

All based on **lattices** and use **bootstrapping** technique

Still more work is needed to make it practical

POTENTIAL APPLICATIONS

Computations on not entirely trusted services (e.g. cloud):

- Encrypt your computational task and send it to a remote server
- The server computes over encrypted data and returns an encrypted result
- Decrypt result

Pipeline processing without revealing intermediate data

Examples:

- Hospitals sharing patient data for medical study
- Similarly for financial, or other sensitive data
- Training ML models directly on encrypted data



CRYPTDB

Similar idea in data processing:

- To query encrypted SQL database without decrypting
- Selected fields can be encrypted
- Practical working prototype system: CryptDB

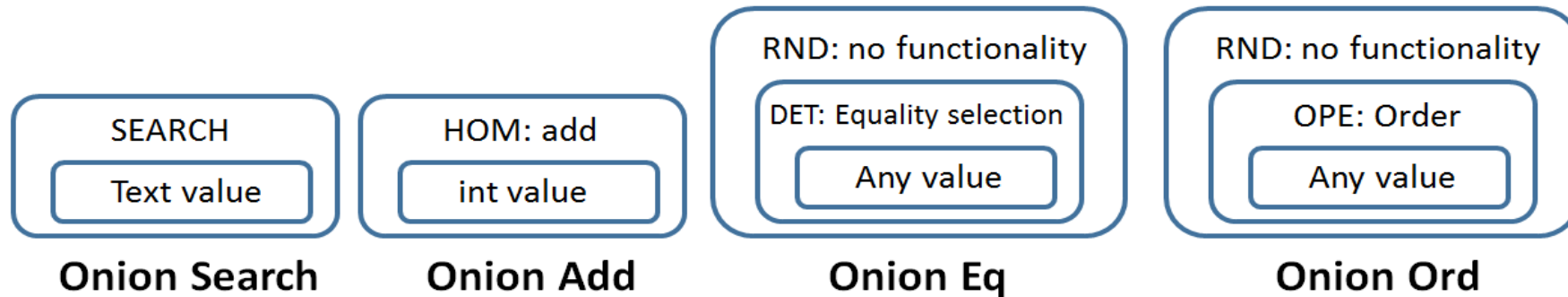
Raluca Ada Popa et al, MIT (2011-):

<http://css.csail.mit.edu/cryptdb/>

Low overhead: reducing throughput 15-25%

Layered classical crypto, practical but limited SQL queries

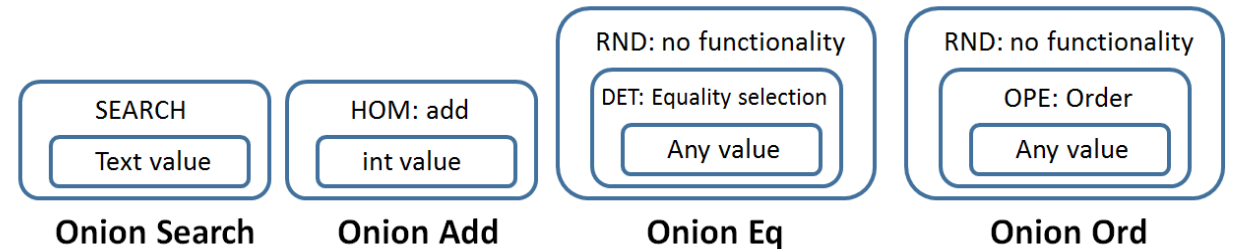
ONION-LAYERED SQL-AWARE ENCRYPTION



All data in CryptDB can be encrypted using several layers of encryption

- Each layer may “release” some information about encrypted value
- Queries cause the system to peel encryption layers until the minimum functionality needed is exposed
- Trade-off: functionality vs information leakage

QUERYING IN CRYPTDB



Before querying, depending on a query:

- some values in the query are encrypted
- encryption layers in the database are adjusted (both steps are done by a proxy)

After the query execution: encrypted results are returned

- The proxy decrypts them and returns to the client the final result

Examples to consider:

```
SELECT * FROM Customers
```

```
SELECT * FROM Customers WHERE Country = 'Mexico' (SQL tutorial at w3schools.com/sql)
```

DEVELOPMENTS HERE IN THE DEPARTMENT

In two PhD projects:

- CryptDB-like approach to graph DBs (Neo4j)
- CryptDB-like approach to document-based DBs (MongoDB)

